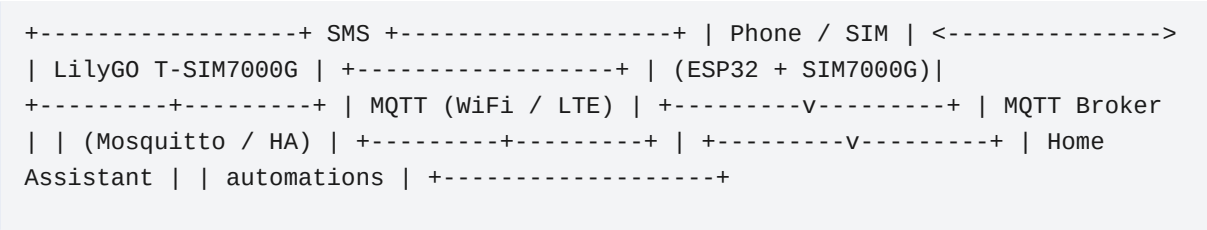


Architecture

Overview



Components

Firmware

Class	File	Responsibility
SmsGateway	SmsGateway.h/.cpp	Orchestration — ties modem and MQTT together
ModemManager	ModemManager.h/.cpp	AT command abstraction over SIM7000G
MqttManager	MqttManager.h/.cpp	MQTT connect / publish / subscribe lifecycle

MQTT Payload Schemas

Incoming SMS (sms/incoming):

```
{ "sender": "+32499000000", "timestamp": "2026-04-30T10:00:00Z", "message": "Hello World" }
```

Send SMS (sms/send):

```
{ "recipient": "+32499000000", "message": "Alert: pH is too low!" }
```

Status (sms/status):

```
{ "state": "online", "rssi": -72, "uptime_s": 3600 }
```

Connectivity

The gateway connects to MQTT over WiFi by default. LTE fallback via SIM7000G can be enabled in `platformio.ini` build flags.